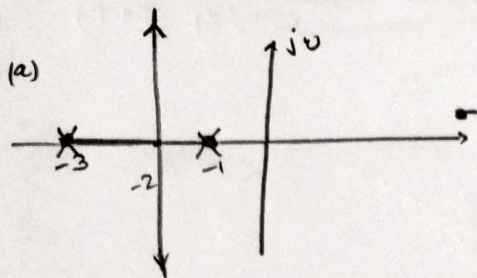
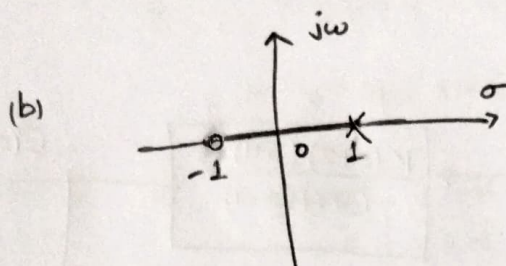


Assignment 4

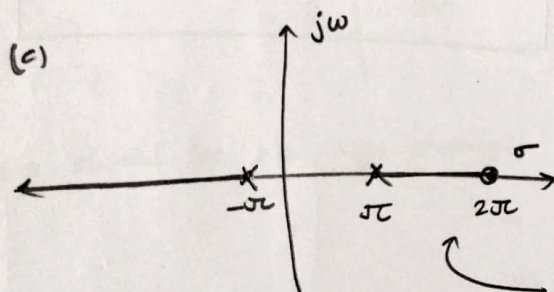
①.



$$G(s) = \frac{1}{(s+1)(s+3)}$$

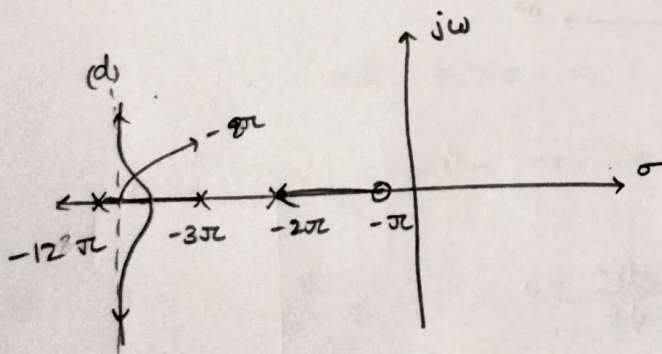


$$G(s) = \frac{(s+1)}{(s-1)}$$



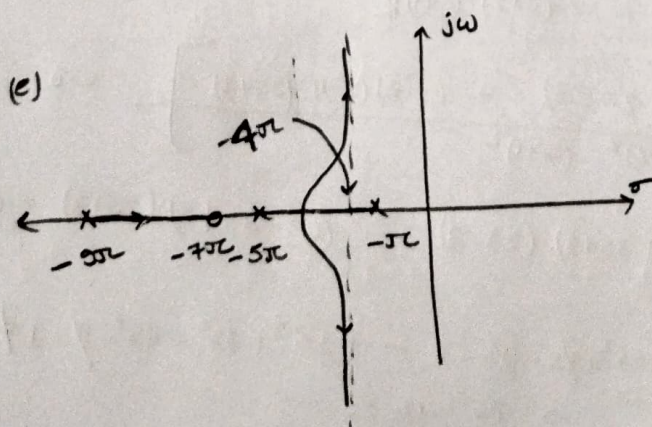
$$G(s) = \frac{(s+2\pi)}{(s+j\pi)(s-j\pi)}$$

Does their direction matter?



$$G(s) = \frac{(s+\pi)}{(s+2\pi)(s+3\pi)(s+8\pi)}$$

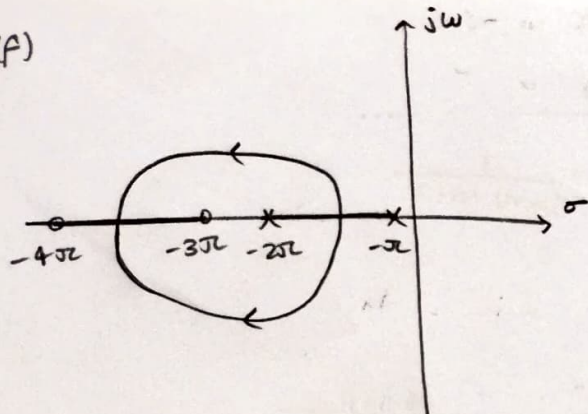
$$\text{asymptote} \Rightarrow \frac{-2\pi - 3\pi - 12\pi + \pi}{2} = -\frac{16\pi}{2} = -8\pi$$



$$G(s) = \frac{(s+7\pi)}{(s+9\pi)(s+\pi)(s+5\pi)}$$

$$\text{asymptote} \Rightarrow \frac{-9\pi - \pi - 5\pi + 7\pi}{2} = -\frac{8\pi}{2} = -4\pi$$

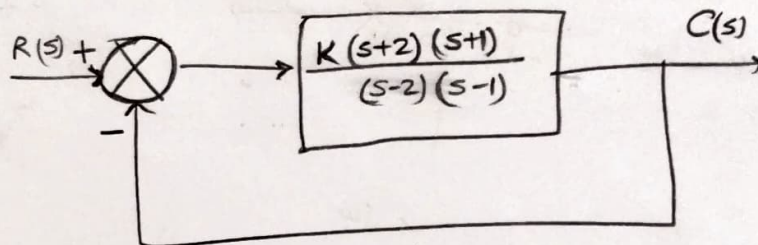
(f)



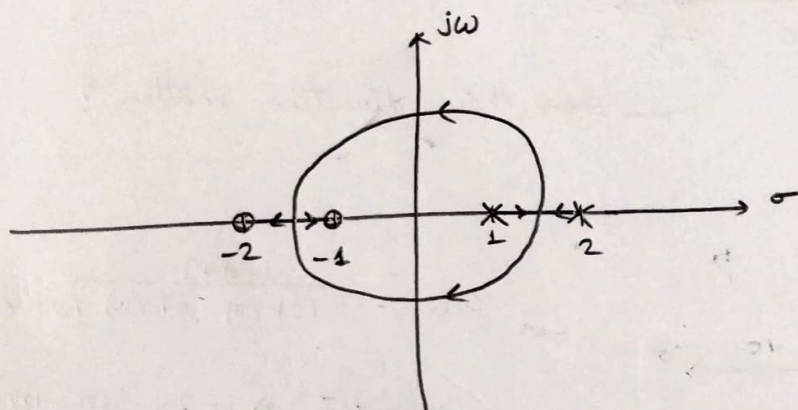
$$G(s) = \frac{(s+4j)(s+3j)}{(s+2j)(s+j)}$$

(2)

(i) System 1



$$G(s) = \frac{(s+2)(s+1)K}{(s-2)(s-1)}$$



(a) for breakaway points, $\frac{dK'}{ds} = 0$

$$\Rightarrow K' = \frac{-1}{G} = \frac{-1(s-2)(s-1)}{(s+2)(s+1)K}$$

$$\frac{dK}{ds} = \frac{-1}{K} \left[\frac{(s+2)(s+1)(2s-3) - (s-2)(s-1)(2s+3)}{(s+2)^2(s+1)^2} \right] = 0$$

$$\Rightarrow (s^2+3s+2)(2s-3) - (s^2-3s+2)(2s+3) = 0$$

$$\Rightarrow 2s^3 - 3s^2 + 6s^2 + 4s - 9s - 6 - (2s^3 + 3s^2 - 6s^2 - 9s + 4s + 6) = 0$$

$$\Rightarrow 6s^2 - 12 = 0$$

$$\Rightarrow s = \pm \sqrt{2}$$

⇒ From graph, Break away point = $+\sqrt{2}$
 and Break-in point = $-\sqrt{2}$

(b) For stability the unity feedback loop

$1+G = 0$ should have roots in LHP

$$\Rightarrow (s-2)(s-1) + K(s+2)(s+1) = 0$$

$$\Rightarrow s^2(1+K) + s(3K-3) + 2K+2 = 0$$

RH table $\Rightarrow$

s^2	$1+K$	$K+1$
s^1	$3K-1$	
s^0	$K+1$	

There should be no sign change $\Rightarrow (K+1)(K-1) > 0$

$$\Rightarrow K^2 - 1 > 0$$

$$\Rightarrow K^2 > 1$$

$$\Rightarrow K > 1 \text{ or } K < -1$$

but $K > 0 \Rightarrow K > 1$

Hence for system to be stable, $K > 1$

$$(d) T(s) = \frac{G}{1+G} = \frac{K(s+2)(s+1)}{K(s+2)(s+1) + (s-2)(s-1)}$$

$$= \frac{K(s^2+3s+2)}{s^2(K+1) + 3s(K-1) + 2(K+1)}$$

$$= \frac{K \frac{(s^2+3s+2)}{(K+1)}}{s^2 + \frac{3s(K-1)}{(K+1)} + 2}$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Why only +ve? (?)

$$\omega_n^2 = 2 \Rightarrow \omega_n = \sqrt{2}$$

$$\text{and } 2\zeta\omega_n = 3 \frac{(K-1)}{(K+1)}$$

$$\Rightarrow 2\sqrt{2}\zeta = 3 \frac{(K-1)}{(K+1)}$$

$$\text{Now } \zeta = 0.707 = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 2 = \frac{3K-3}{K+1}$$

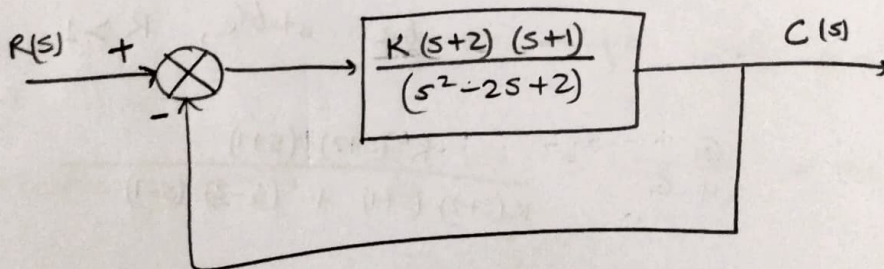
$$\Rightarrow 2K+2 = 3K-3$$

$$\Rightarrow K=5$$

$\Rightarrow K=5$ for $\zeta=0.707$ with stable system.

$$\begin{aligned} \text{(c)} \quad \zeta = 1 & \Rightarrow 2\sqrt{2}(1) = \frac{3K-3}{K+1} & \Rightarrow 2\sqrt{2}K + 2\sqrt{2} = 3K-3 \\ & \Rightarrow K(3-2\sqrt{2}) = 2\sqrt{2}+3 \\ & \Rightarrow K = \frac{2\sqrt{2}+3}{3-2\sqrt{2}} \end{aligned}$$

(ii) System 2

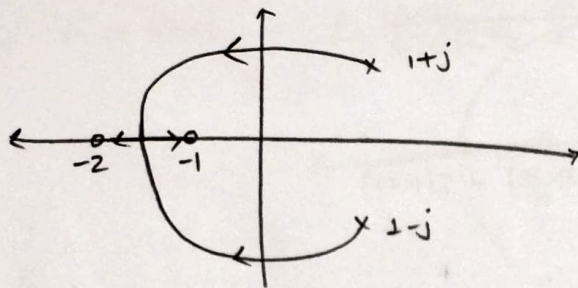


$$T(s) = \frac{G(s)}{1+G(s)}$$

$$G(s) = \frac{K(s+2)(s+1)}{(s^2-2s+2)}$$

$$s^2 - 2s + 2 = 0$$

$$\Rightarrow s = \frac{2 \pm \sqrt{4-4 \times 2}}{2} = 1 \pm \sqrt{1-2} = 1 \pm j$$



(a) Breakaway points $\Rightarrow K' = \frac{-1}{G(s)} \quad \& \quad \frac{dK'}{ds} = 0$

$$\Rightarrow K' = -\frac{1}{K} \frac{(s^2 - 2s + 2)}{(s+2)(s+1)}$$

$$\frac{dK'}{ds} = -\frac{1}{K} \left[\frac{(2s-2)(s+2)(s+1) - (s^2-2s+2)(2s+3)}{(s+2)(s+1)^2} \right] = 0$$

$$\Rightarrow (2s-2)(s^2+3s+2) - (s^2-2s+2)(2s+3) = 0$$

$$\Rightarrow \cancel{2s^3} + \cancel{6s^2} + \cancel{4s} - \cancel{2s^3} - \cancel{6s^2} - 4 - \left(\cancel{2s^3} + \cancel{3s^2} - \cancel{4s^2} - \cancel{6s} + \cancel{4s} + 6 \right) = 0$$

$$\Rightarrow 5s^2 - 10 = 0$$

$$\Rightarrow s = \pm \sqrt{2}$$

$$\Rightarrow \boxed{s = -\sqrt{2} \text{ for Break-in points}}$$

(b) $1 + G(s) = 0$'s roots are in RHP

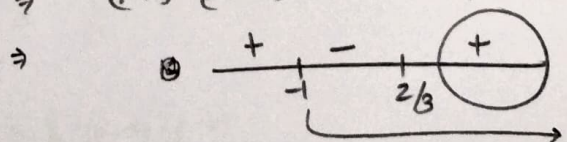
$$(s^2 - 2s + 2) + K(s+2)(s+1) = 0$$

$$s^2(K+1) + s(3K-2) + 2(K+1) = 0$$

s^2	$K+1$	$K+1$
s^1	$3K-2$	
s^0	$2(K+1)$	

No sign change

$$\Rightarrow (K+1)(3K-2) > 0 \quad \& \quad K > -1$$



$$\Rightarrow K > 2/3$$

So to keep system stable

$$\boxed{K > \frac{2}{3}}$$

$$(d) \quad T(s) = \frac{K(s+2)(s+1)}{K(s+2)(s+1) + s^2 - 2s + 2}$$

$$= \frac{K(s+2)(s+1)}{s^2(K+1) + s(3K-2) + 2(K+1)}$$

$$= \frac{K(s+2)}{s^2 + s\left(\frac{3K-2}{K+1}\right) + 2} = \frac{C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n^2 = 2 \Rightarrow \omega_n = \sqrt{2}$$

$$\therefore 2\zeta\omega_n = \frac{3K-2}{K+1}$$

$$2\sqrt{2}\zeta = \frac{3K-2}{K+1}$$

$$\zeta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 2K+2 = 3K-2$$

$$\Rightarrow \boxed{K=4}$$

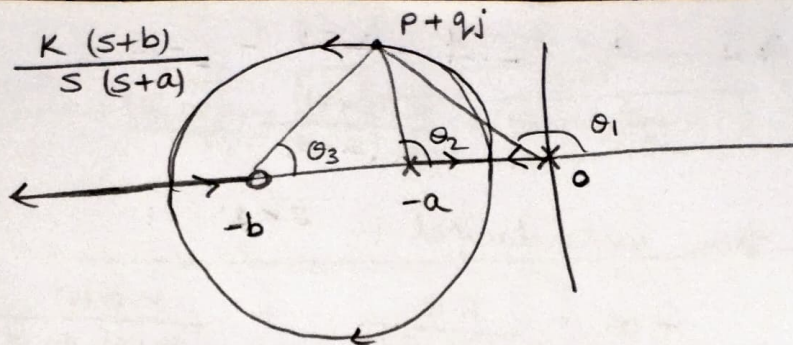
$$(c) \quad \zeta = 1 \quad 2\sqrt{2} = \frac{3K-2}{K+1}$$

$$(3-2\sqrt{2})K = 2\sqrt{2}+2$$

$$\Rightarrow \boxed{K = \frac{2\sqrt{2}+2}{3-2\sqrt{2}}}$$

3.

$$G(s)H(s) = \frac{K(s+b)}{s(s+a)}$$



$$-\theta_1 - \theta_2 + \theta_3 = (2n+1)\pi$$

$$-\left(\pi - \tan^{-1}\left(\frac{q}{p}\right)\right) - \left(\pi - \tan^{-1}\left(\frac{q}{p-a}\right)\right) + \tan^{-1}\left(\frac{q}{-p+b}\right) = (2n+1)\pi$$

$$\Rightarrow -2\pi + \tan^{-1}\left(\frac{q}{p}\right) + \tan^{-1}\left(\frac{q}{p-a}\right) + \tan^{-1}\left(\frac{q}{-p+b}\right) = (2n+1)\pi$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{q}{p-a} + \frac{q}{-p+b}}{1 - \frac{q^2}{(p-a)(p+b)}}\right) + \tan^{-1}\left(\frac{q}{p}\right) = (2n+1)\pi$$

$$\Rightarrow \tan^{-1}\left(\frac{q(b-a)}{(p-a)(b-p) - q^2}\right) + \tan^{-1}\left(\frac{q}{p}\right) = (2n+1)\pi$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{q(b-a)}{(p-a)(b-p) - q^2} + \frac{q}{p}}{1 - \frac{q(b-a)q}{((p-a)(b-p) - q^2)p}}\right) = (2n+1)\pi$$

taking tan on both sides $\Rightarrow$

$$\frac{q(b-a)}{(p-a)(b-p) - q^2} + \frac{q}{p} = 0$$

$$\frac{b-a}{(p-a)(b-p) - q^2} = -\frac{1}{p}$$

$$\begin{aligned} \frac{pb - pa}{p(b-a)} &= q^2 + (p-a)(p-b) \\ &= q^2 + p^2 - (a+b)p + ab \end{aligned}$$

$$\Rightarrow q^2 + p^2 - 2bp + ab = 0$$

$$\Rightarrow (p-b)^2 + q^2 = (\sqrt{b^2 - ab})^2$$

$\rightarrow$ a circle

for $a = 4$ & $b = 6$

$$G(s)H(s) = \frac{K(s+6)}{s(s+4)}$$

For underdamped $\zeta < 1$

$$T(s) = \frac{G(s)}{1 + G(s)} = \frac{K(s+6)}{K(s+6) + s(s+4)} = \frac{K(s+6)}{s^2 + s(K+4) + 6K}$$

$$\omega_n^2 = 6K \Rightarrow \omega_n = \sqrt{6K}$$

$$\Rightarrow 2\zeta\omega_n = K+4$$

$$\Rightarrow 2\sqrt{6K}\zeta = K+4$$

$$\Rightarrow \zeta = \frac{K+4}{\sqrt{24K}} < 1$$

$$\Rightarrow \frac{K+4 - \sqrt{24K}}{\sqrt{24K}} < 0$$

$$\Rightarrow K^2 + 16 + 8K < 24K$$

$$\Rightarrow K^2 - 16K + 16 < 0$$

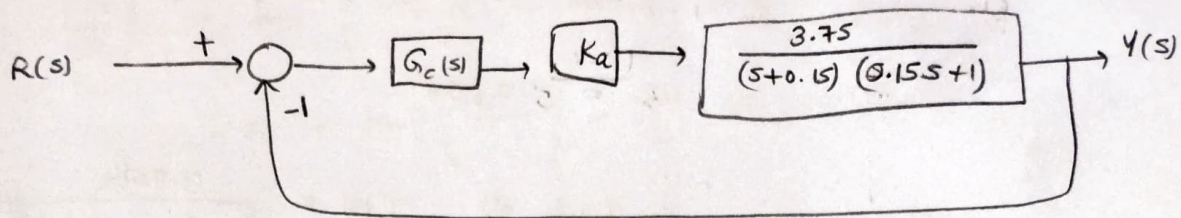
$$\Rightarrow (K-8)^2 < 48$$

$$\Rightarrow K < 8 - 4\sqrt{3} \quad \text{+} \quad K > 8 + 4\sqrt{3}$$

$$\Rightarrow \text{Hence } \boxed{K > 8 + 4\sqrt{3}}$$

$$\text{But } \sqrt{24K} > 0 \\ K > 0$$

④



$$\therefore 0.5 = 10\% \Rightarrow \zeta = \frac{-\ln(0.05)}{\sqrt{\ln^2(0.05) + \pi^2}}$$

$$= \frac{-\ln(0.1)}{\sqrt{\ln^2(0.1) + \pi^2}} = 0.5911$$

$$T_s < 1.5 \Rightarrow \frac{4}{\zeta \omega_n} < 1.5$$

$$\Rightarrow \zeta \omega_n > \frac{8}{3}$$

$e_{ss} = 0$ for a step input $\Rightarrow$ insert pole at origin $\Rightarrow K_p = \infty$

But to compensate for the pole, we need another zero at '-a':

$$\Rightarrow \text{asymptote} = \frac{-\frac{207}{3} - 0.15 - 0 + a}{2}$$

$$= -3.408 + \frac{a}{2} \rightarrow \text{should be left of } -\frac{8}{3}$$

$\swarrow$
 $\zeta \omega_n$

$$\Rightarrow -3.408 + \frac{a}{2} < -\frac{8}{3}$$

$$\Rightarrow a < 1.4826$$

$$\text{but } a \in (-0.15, 0)$$

$$\Rightarrow \text{let } a = +0.1$$

$$\Rightarrow G_c(s) = \frac{(s+0.1)}{s} = 1 + \frac{0.1}{s}$$

$$\Rightarrow G'(s) = \frac{K_a(3.75)(s+0.1)}{s(s+0.15)(0.15s+1)}$$

asymptotes at $-3.408 = -\omega_n \delta = -\omega_n 0.591$

$$\Rightarrow \omega_n = 5.7655$$

Closed loop feedback $\Rightarrow T(s) = \frac{3.75K_c}{3.75K_c + 0.15s^2 + (0.15)^2s + 5s + 0.15}$

Now $\omega_n^2 = \frac{3.75K_c + 0.15}{0.15}$

$$\Rightarrow K_c = 1.29 \quad \& \quad T_s \text{ also } < 1.5 \text{ s}$$

⑤

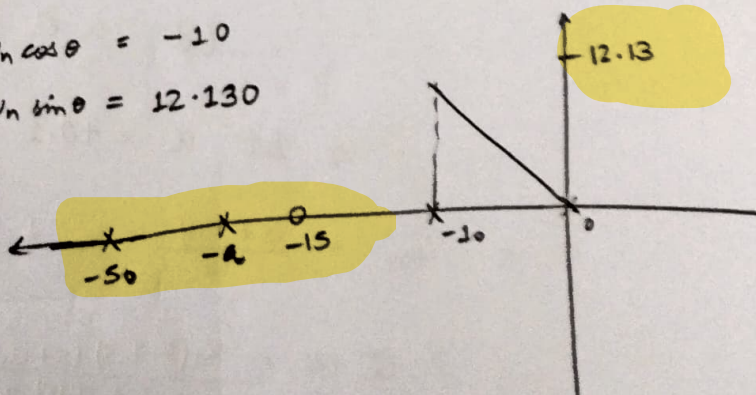
$$G(s) = \frac{K}{s(s+10)(s+50)}$$

$\therefore 0.05 = 7.5\%$ $\Rightarrow \zeta = \frac{-\ln(0.05)}{\sqrt{\ln^2(0.05) + \pi^2}} = \frac{-\ln(0.075)}{\sqrt{\ln^2(0.075) + \pi^2}} = 0.636$

$T_s = \frac{4}{\zeta \omega_n} < 400 \times 10^{-3} \Rightarrow \zeta \omega_n = 10$
 $\Rightarrow \omega_n > 15.72$

Compensator zero at $s = -15$
 let pole at $-a$

$-\omega_n \cos \theta = -10$
 $\omega_n \sin \theta = 12.130$



$$\tan^{-1}\left(\frac{12 \cdot 13}{5}\right) - \tan^{-1}\left(\frac{12 \cdot 13}{a-10}\right) - \tan^{-1}\left(\frac{12 \cdot 13}{40}\right) - \pi + \tan^{-1}\left(\frac{12 \cdot 13}{10}\right) - \frac{\pi}{2} = (2n+1)\pi$$

$$\Rightarrow \frac{12 \cdot 13}{a-10} = 0.2$$

$$\Rightarrow 0.2a = 14 \cdot 13$$

$$\Rightarrow a = 70.65$$

$$\Rightarrow G'(s) = \frac{(s+15)}{s(s+10)(s+50)(s+70.65)}$$

$$\text{at } (-10, 12 \cdot 13) \quad 1 + KG'(s) = 0$$

$$K = \frac{-1}{G'(s)}$$

$$\Rightarrow K = 37575.35 + 54.84j$$

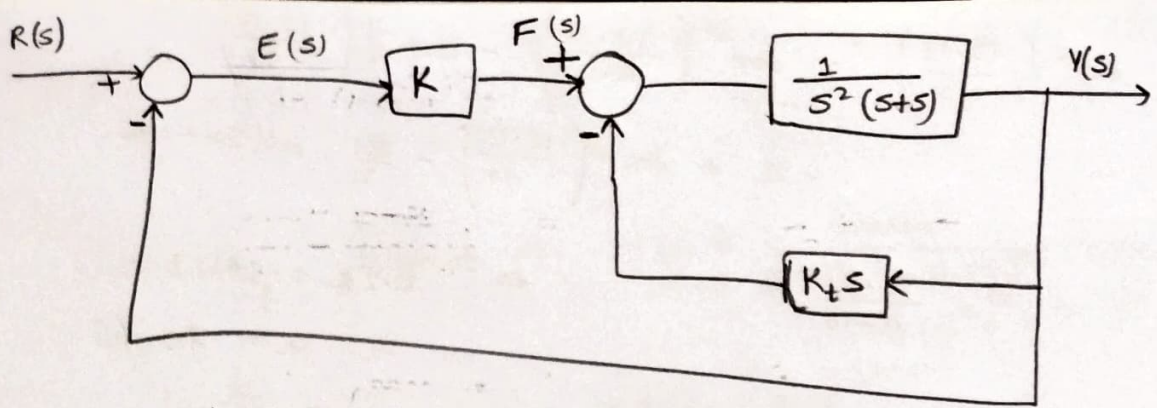
$$|K| \approx 37575.39$$

$$K_v = \lim_{s \rightarrow 0} s G(s)$$

$$= \lim_{s \rightarrow 0} \frac{(s+15)(37575.39)}{(s+10)(s+50)(s+70.65)}$$

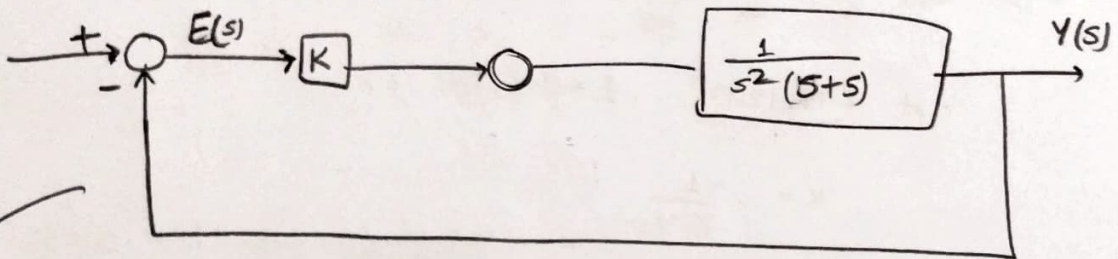
$$= 15.95$$

7



(a)

$K_t = 0$ $K > 0$



$$E(s) = R(s) - Y(s)$$

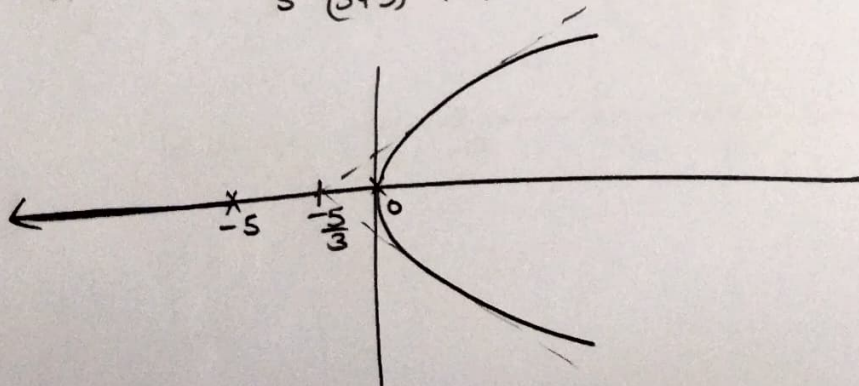
$$F(s) = K E(s)$$

$$\frac{(F(s) - K_t s Y(s))}{s^2 (s+5)} = Y(s)$$

$$\Rightarrow \frac{F(s)}{s^2 (s+5)} = \frac{Y(s) \left(1 + \frac{K_t s}{s^2 (s+5)} \right)}{s^2 (s+5)}$$

$$\Rightarrow \frac{F(s)}{Y(s)} = \frac{1}{s \left(s(s+5) + K_t \right)} \rightarrow \text{general equation}$$

$$T(s) = \frac{K}{s^2 (s+5) + K}$$



b)

$$\frac{K(R(s) - Y(s))}{Y(s)} = \frac{1}{s(s(s+5) + K_t)}$$

$$\Rightarrow K \frac{R(s)}{Y(s)} - K = \frac{1}{s(s(s+5) + K_t)}$$

$$\Rightarrow \frac{R(s)}{Y(s)} = \frac{1}{Ks(s(s+5) + K_t)} + 1$$

$$K = 10$$

$$K_t > 0$$

$$\Rightarrow T(s) = \frac{1}{10s(s(s+5) + K_t)} + 1$$

or

$$G(s) = \frac{K}{s(s(s+5) + K_t)}$$

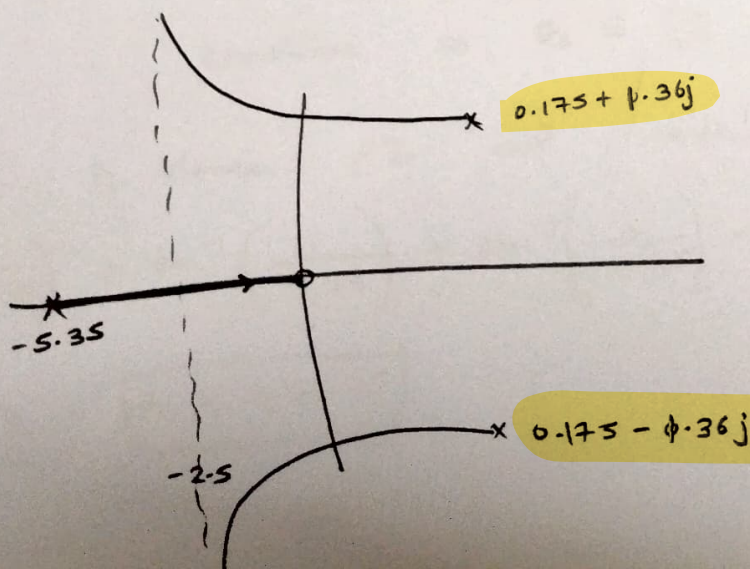
$$\Rightarrow T(s) = \frac{10}{s^3 + 5s^2 + K_t s + 10}$$

$$= \frac{10 / (s^3 + 5s^2 + 10)}{1 + \frac{K_t s}{s^3 + 5s^2 + 10}}$$

etc plot.

$$\text{Poles} = -5.35, 0.175 + 1.36j, 0.175 - 1.36j$$

$$\text{asymptote} = \frac{-5.35 + 2 \times 0.175}{2} = -2.5$$



(9)

$$G(s) = \frac{K(s+a)}{s(s+3)(s+6)}$$

$$T(s) = \frac{K(s+a)}{K(s+a) + s(s+3)(s+6)} = \frac{K(s+a)}{s^3 + 9s^2 + (18+K)s + Ka}$$

pair of poles = $-1 + j100, -1 - j100$

$\Rightarrow$ sum of poles = -9

$$-9 = -1 + \cancel{100j} - 1 - \cancel{100j} + m$$

$\Rightarrow m = -7 = \text{third root}$

$$18+K = -7(-1+100j) - 7(-1-100j) + (-1-100j)(-1+100j)$$

$$= 7 - 700j + 7 + 700j + 1 + (100)^2$$

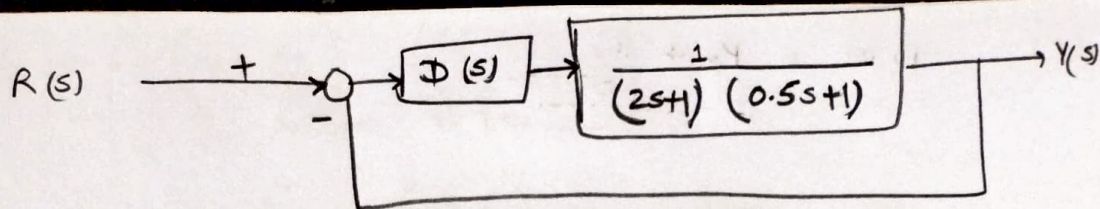
$\Rightarrow K = 9997$

$$Ka = 7(1 + (100)^2) \Rightarrow$$

$$a = \frac{70007}{9997} = 7.003$$

$7 > 4 \times -1 \Rightarrow$ we can approximate it with a 2nd order closed loop TF.

(10)



$$\delta = 0.6$$

$$T = \frac{1}{f_{wn}} = \frac{1}{0.75}$$

$$f_{wn} = 0.75$$

$$e_{ss} = 0 \Rightarrow K_p = \infty \text{ for step input}$$

$$\Rightarrow \text{at least one pole at origin}$$

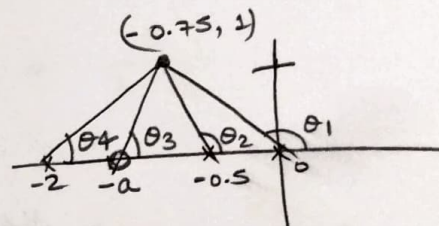
(a) Proportional control K_p won't give a pole at origin and thus is not adequate.

(b) PI control will work because $\Rightarrow$

Locus should pass through $(-f_{wn}, +\sqrt{1-\delta^2} f_{wn}) = (-0.75, 1)$

Let PI be $\Rightarrow \frac{K(s+a)}{s}$

$$\Rightarrow G(s) = \frac{K(s+a)}{s(2s+1)(0.5s+1)}$$



$$\text{sum of } \theta_3 - (\pi - \theta_1) - (\pi - \theta_2) - \theta_4 = (2n+1)\pi$$

Now for fixed $\theta_1, \theta_2, \theta_4$, we will get a value of a for which θ_3 will satisfy the above condition as $\theta_3 \in (0, \theta_1)$

$\Rightarrow$ Hence PI will work.

$$\tan^{-1}\left(\frac{1}{0.75-a}\right) + \tan^{-1}\left(\frac{1}{0.75}\right) + \tan^{-1}\left(\frac{1}{0.25}\right) - \tan^{-1}\left(\frac{1}{1.25}\right) = (2n+1)\pi$$

$$\Rightarrow a = 0.7575$$

$$(c) \quad D(s) = K_p + \frac{K_I}{s} \quad \rightarrow \text{PI control}$$

$$\Rightarrow \quad D(s) = \frac{K_p \left(s + \frac{K_I}{K_p} \right)}{s}$$

from (b), we get

$$a = 0.7575 = \frac{K_I}{K_p}$$

now, for $s = -0.75 + 1j$ $K = \frac{-1}{G(s)}$

$$\Rightarrow K = \frac{(-0.75 + j) + 0.7575}{(-0.75 + j)(-1.5 + 2j + 1) \left(\frac{1}{2}(-0.75 + j) + 1 \right)}$$

$$= 2.062 - 1.5 \times 10^{-4} j$$

$$|K| = 2$$

$$\Rightarrow K_p = K = 2 \quad \Rightarrow K_I = 0.7575 K_p$$

$$\Rightarrow K_I = 1.515 \quad \text{if } K_p = 2$$

⑧

$$G(s) = \left(K_p + \frac{K_I}{s} \right) \left(\frac{48500}{s^2 + 2.89s} \right)$$

(a) $e_{ss} = 2 \rightarrow$ for parabolic input $= \frac{1}{K_a}$

$$\Rightarrow K_a = \frac{1}{2} = \lim_{s \rightarrow 0} s^2 G(s) = \frac{1}{2}$$

$$\Rightarrow \lim_{s \rightarrow 0} s^2 \left(\frac{K_p + \frac{K_I}{s}}{s} \right) \left(\frac{48500}{s^2 + 2.89s} \right) = \frac{1}{2}$$

$$\Rightarrow \lim_{s \rightarrow 0} (s^2 K_p + s K_I) \left(\frac{48500}{s^2 + 2.89s} \right) = \frac{1}{2}$$

$$\Rightarrow \lim_{s \rightarrow 0} (s K_p + K_I) \left(\frac{48500}{s + 2.89} \right) = \frac{1}{2}$$

$$\Rightarrow \frac{K_I (48500)}{2.89} = \frac{1}{2} \Rightarrow K_I = 2.979 \times 10^{-5}$$

(b) $T(s) = \frac{\left(K_p + \frac{K_I}{s} \right) 48500}{s^2 + 2.89s + \left(K_p + \frac{K_I}{s} \right) 48500} = \frac{G}{1+G}$

$$= \frac{\left(K_p + \frac{K_I}{s} \right) 48500}{s^2 + 2.89s + \left(K_p + \frac{2.979 \times 10^{-5}}{s} \right) 48500}$$

$$= \frac{(s K_p + K_I) 48500}{s^3 + 2.89s^2 + K_p s (48500) + 1.44}$$

$$= \frac{(s K_p + K_I) 48500}{s^3 + 2.89s^2 + 1.44}$$

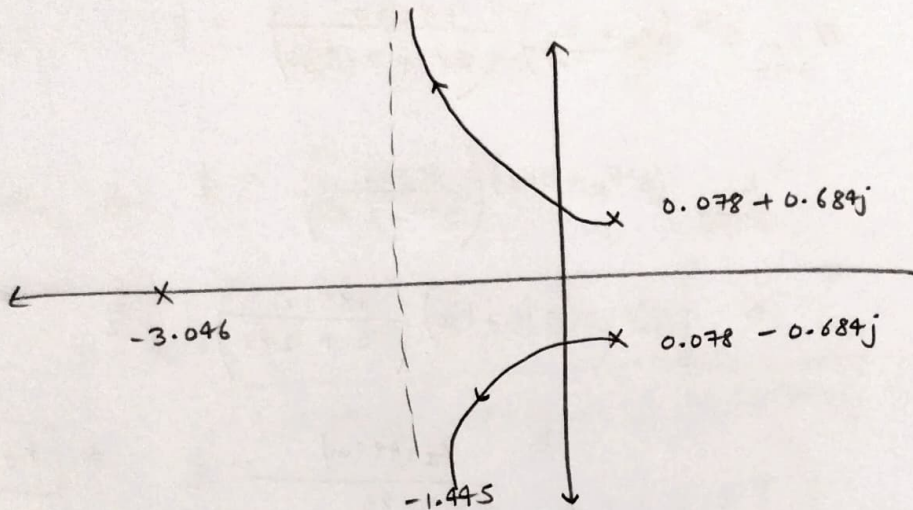
$$= \frac{1 + K_p \left(\frac{s (48500)}{s^3 + 2.89s^2 + 1.44} \right)}{1 + K_p G(s)}$$

$$= \frac{C}{1 + K_p G(s)}$$

plot this

$$\Rightarrow G(s) = \frac{48500s}{(s+3.046)(s-0.078-0.684j)(s-0.078+0.684j)}$$

$$\Rightarrow \text{Asymptote point} = \frac{-3.046 + 2 \times 0.078 + 0}{2} = 1.445$$



(C) Real pole at -1. $\rightarrow$ I will be wrong

$$K_p = \frac{-1}{G(1)} = 6.876 \times 10^{-5}$$

$$1 + G(s)K_p = 0 \Rightarrow s^3 + 2.895s^2 + 1.445 \times 3.335s = 0$$

$$\Rightarrow s = -1, -0.94 \pm 0.7399j$$

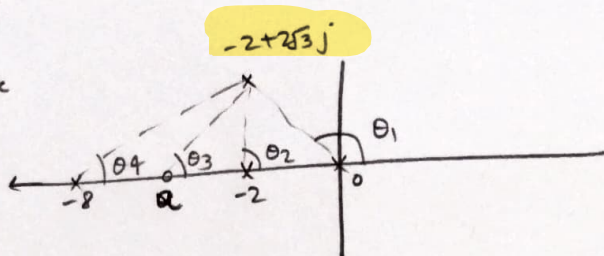
⑥

$$G(s) = \frac{10}{s(s+2)(s+8)}$$

Poles at $s = -2 \pm 2\sqrt{3}j$

applying PD compensator

$$G'(s) = \frac{K(s+a)10}{s(s+2)(s+8)}$$



$$\theta_3 - \theta_1 - \theta_2 - \theta_4 = (2n+1)\pi$$

$$\Rightarrow \tan^{-1}\left(\frac{2\sqrt{3}}{2-a}\right) - \left(\pi - \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) + \frac{\pi}{2} + \tan^{-1}\left(\frac{2\sqrt{3}}{6}\right)\right) = (2n+1)\pi$$

$\swarrow 60^\circ$
 $\nearrow 30^\circ$

$$\Rightarrow \tan^{-1}\left(\frac{2\sqrt{3}}{2-a}\right) - 240 = -180^\circ$$

$$\Rightarrow \tan^{-1}\left(\frac{2\sqrt{3}}{2-a}\right) = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow 2\sqrt{3} = 2\sqrt{3} - \sqrt{3}a \rightarrow a = 0 \rightarrow \text{not possible}$$

or

$$2\sqrt{3} = -2\sqrt{3} + \sqrt{3}a \rightarrow a = 4 \rightarrow \text{possible and right}$$

$$\Rightarrow K = \frac{-1}{G(s)} = \frac{s(s+2)(s+8)}{10(s+4)} \text{ at } s = -2 \pm 2\sqrt{3}j$$

$$\Rightarrow K = 2.4$$

Now applying a lag compensator for $K_v \Rightarrow \left(\frac{s+\alpha}{s+\beta}\right) G'(s)$

$$K_v = \lim_{s \rightarrow 0} s G(s) = \frac{10 \times 2.4 \times 4 \times \alpha/\beta}{8 \times 2} = 80$$

$$\Rightarrow \frac{\alpha}{\beta} = 13.33$$

Now taking very small $\alpha, \beta \Rightarrow \beta = 10^{-5} \Rightarrow \alpha = 13.33 \times 10^{-5}$

$$\Rightarrow \text{Final compensator} = \frac{2.4(s+4)(s+13.33 \times 10^{-5})}{(s+10^{-5})}$$